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Sex disparities in outcome of medication-assisted therapy of opioid use disorder: Nationally representative outpatient clinic data

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Abstract

Question: The opioid epidemic causes massive morbidity, and males have substantially greater overdose mortality rates than females. It is unclear whether there are sex-related disparities at different stages in the trajectory of opioid use disorders (OUD), from large samples in the community.

Goal: To determine sex disparities in non-medical opioid use (NMOU) at the end of treatment with medications for opioid use disorder (MOUD), using national data.

Design: Observational study of outpatient MOUD programs in the “Treatment episode data set-discharges” (TEDS-D) for 2019.

Participants: Persons aged ≥ 18 in their first treatment episode, in outpatient MOUD-based therapy for heroin or other opioids ($N=10,065$). The binary outcome was presence/absence of NMOU in the month prior to discharge.

Results: In univariate analyses, males had higher odds of NMOU compared to females (odds ratio=1.22; $p=6.84 \times 10^{-5}$ after Bonferroni correction). A multivariable logistic regression

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detected a relatively small male>female odds ratio of 1.14 ($p=0.0039$), surviving adjustment for demographic variables and social determinants of health. Several specific conditions were revealed in which males had greater odds of NMOU compared to females (e.g., if they were in the white racial category, and were not of hispanic ethnicity). Also, using by smoking, inhalation or injection routes (versus oral) was associated with greater odds of NMOU, irrespective of sex.

Conclusions: This national community sample shows that males overall have greater odds of NMOU in their first treatment episode with MOUD, a potential indicator of more unfavorable outcomes. Further analyses should examine the underpinnings of this disparity, including clinical severity features.

Keywords

opioid; methadone; buprenorphine; fentanyl; medication-assisted treatment; medications for opioid use disorders; real world data

1. Introduction

The opioid epidemic causes massive health consequences in the U.S. [1,2], and we recently reported approximately 2.5-fold greater opioid-induced overdose mortality rates in males versus females, even after adjusting for sex-specific frequency of non-medical opioid use (NMOU) [1]. Therefore, this sex disparity was not simply due to differing levels of overall opioid use, and may reflect multiple biological, neurocognitive and social-behavioral factors, as well as response to treatment [3–7]. Understanding the bases for these differences could inform the development of personalized sex or gender-specific interventions, and decrease morbidity and mortality [6,8–10].

Several studies have compared different stages in the trajectory of opioid use disorders (OUD) in males and females, as well as response to medications for opioid use disorder (MOUD) [10–12]. Overall, it is difficult to draw overall conclusions across studies on MOUD outcomes, due to differences across time, cohort, study methodology, and environment [11–13]. Thus, some studies have reported that females have greater treatment retention for MOUD compared to males [10,14–17], but others show greater retention in males [18,19], or no sex difference [20,21].

It has been noted that some formal research studies in SUD may not be representative of community clinical settings, due to relative lack of inclusion based on socio-demographic factors and social determinants of health, or due to the types of institutions represented [22–25]. Therefore, data obtained in the course of clinical practice in large and diverse samples (outside of formal research studies) can be of value [26–28]. Intriguingly, one such nationally representative study detected that females in MOUD had 1.1 greater odds of staying in treatment for at least 6 months, compared to males [16]. However, duration of treatment in MOUD can differ because of environmental and program-level features, in addition to client-level clinical factors [16,29]. Other outcomes such as “treatment completion” or “termination” may also be influenced by program-level regulations, and social determinants of health [30–33]. Overall, a major favorable clinical outcome of this type of therapy is decreased NMOU [8,9,34]. Therefore, this study examined sex differences

in the frequency of NMOU in the month prior to discharge [35,36], in a national outpatient community clinic sample, adjusting for length of stay in treatment, as well as demographic variables and social determinants of health [37–39].

2. Methods

2.1 Data Source:

This study used the Treatment Episode Data Set-Discharges (TEDS-D), from 2019 (<https://www.datafiles.samhsa.gov/>), the most recent data currently available prior to the disruptions of the COVID-19 pandemic [40]. The TEDS-D is composed of individual episodes from substance use treatment facilities that receive public funding; data includes the continental US, excluding Oregon, Washington state and West Virginia. The TEDS-D codebook [41] provides information on the sampled population, major stratification features and available variables that each clinic enters into the system. Further granular data on individual clinic procedures is not available. Data extraction and variable coding for this study are described below.

2.2 Outcome variable:

The outcome variable was *NMOU in the month prior to discharge* (TEDS-D variable `FREQ1_D`), recoded as a binary variable, combining “some use” and “daily use”, versus “no use”. As all other variables in the TEDS-data set, this is based on clinic records which can have differing local standards (e.g., could include self-report, urine tests, or a combination thereof). The “some use” and “daily use” categories were combined, in order to capture all cases where NMOU occurred; this can be a potential limitation.

2.3 Inclusion Criteria:

A) First episodes of treatment (variable `NOPRIOR=0`). This criterion avoids confounding due to potentially examining several episodes in the same person, or due to prior treatment history [42,43]. **B)** Episodes in which an opioid is the primary substance (i.e., heroin, non-prescription methadone, or “other opiates and synthetics”) at admission and discharge (i.e., variables `SUB1` and `SUB1_D=5,6, or 7`, respectively). **C)** Episodes with MOUD in the treatment plan (variable `METHUSE=1`; principally methadone and buprenorphine, but also naltrexone) [9,44,45]. As a limitation, the “METHUSE” variable pools these medications, not allowing further stratification, and does not provide information on treatment duration and adherence. **D)** Service in an ambulatory (outpatient) setting (variable `SERVICES=6 or 7`) [8,9,44]. **E)** Episodes with a length of stay (variable `LOS`) of ≥ 30 days, to focus on those who had at least initial treatment engagement [16,46]. **F)** Episodes in census divisions 1–9 (i.e., continental U.S.).

2.4 Exclusion criteria:

A) Age < 18 , due to low prevalence. **B)** Episodes terminated because the person had died (cause of death is unavailable; variable `REASON=5`), or became incarcerated (variable `REASON=6`). **C)** Episodes in which the route of non-medical opioid use was “other” (i.e., variable `ROUTE1=5`), due to low prevalence. **D) Missing data:** If there were missing data in any of these variables, the episode was excluded from analysis (episode-wise deletion).

2.5 Demographic variables:

In addition to sex, several variables were examined: Length of stay in treatment (variable LOS), coded in bins denoting increasing duration (31–45, 46–60, 61–90, 91–180, 181–365, ≥ 366 days). Route of NMOU (variable ROUTE1) was coded as four categories: oral, smoking, inhalation or injection. Age at admission was recoded as 18–29, 30–39, 40–49, ≥ 50 age bins. Housing at discharge was coded as: independent housing, dependent (e.g., supportive) housing, and homeless (unhoused). Employment at discharge was coded as: employed full time, employed part-time, “not in the labor force” and unemployed. Racial background was studied as the two main available categories (African-American/black and white). We excluded persons in other specific racial groups, due to their relatively low prevalence in this dataset [47–49]. Psychiatric comorbidity was coded as a binary variable (present/absent), without further stratification available. Hispanic/latino ethnicity was re-coded as a binary variable (any hispanic/latino versus non-hispanic/latino). Referral source was re-coded as a binary variable, with court/criminal justice referrals versus all others. Census Divisions were also examined, for geographic control.

2.6 Data analyses:

Analyses were carried out with GraphPad Prism (V. 10) and TIBCO Statistica Data Science Workbench. The alpha-level of significance was $p=0.05$.

2.6.1 Univariate Analyses—Demographics were examined with χ^2 analyses across sex. The relative frequency of the outcome (use vs no use of non-medical opioids) in males and females was also examined in χ^2 2X2 contingency analyses. The p-values for these contingency analyses were Bonferroni-corrected, and male/female odds ratios are shown for those that survive correction (using the Baptista-Pike method). 95%CI for the proportions of males and females with the outcome were calculated with the Wilson-Brown method [50]. Data are also presented as % of males and females with NMOU, and male-female difference.

2.6.2 Multivariable analysis—A multivariable logistic regression [51,52] using maximum likelihood estimation was carried out for the binary outcome of NMOU, versus no use. Sex and all the variables above were entered in the regression. Adjusted Beta-parameters and odds ratios of the outcome are reported. Regression performance was examined with receiver operating characteristic (ROC) analysis, Nagelkerke R^2 and Hosmer-Lemeshow test [53,54].

2.7 Sensitivity Analyses:

a) Exploration of interaction terms between sex and other relevant variables in the multivariable regression model, **b)** Examination of the subset whose primary drug was heroin (as opposed the combined group with heroin, non-medical methadone and “other opiates and synthetics”), and **c)** Examination of major poly-drug use patterns, with the subset who had an opioid as primary substance *and* either cocaine or methamphetamine as a secondary substance (using variable SUB2).

3. Results:

After the inclusion and exclusion criteria were satisfied, and deletion of episodes with missing data in any of the studied variables, $n=10,065$ episodes were available for analysis.

3.1 Sample demographics by sex:

The number of males and females for each condition is in Supplementary Table S1, examined with χ^2 contingency analyses. Among sex disparities observed, there was a smaller relative frequency of males versus females with a psychiatric comorbidity.

3.2.1 Univariate association of sex with the outcome (NMOU): Full data are shown in Supplement Table S2. Overall, 62.4% of males and 56.4% of females had NMOU. The 2X2 contingency analysis (male/female X use/no use) was significant ($\chi^2[1]=22.89$; uncorrected $p=1.71 \times 10^{-6}$; Bonferroni-corrected $p=6.84 \times 10^{-5}$) with a male/female unadjusted odds ratio of =1.22 (95%CI: 1.12–1.32) (Table S2; Figure 1).

3.2.2 Univariate association of demographic and clinical variables with NMOU, by sex: Full data are shown in Supplement Table S2. Contingency analyses examined sex-associated outcomes. Figure 1A shows %non-medical opioid use in males and females; Figure 1B shows the Male-Female difference. After correction for multiple comparisons, males had greater odds of NMOU compared to females in the following conditions: **a)** if they were in independent housing, **b)** employed part-time or unemployed, **c)** white racial category or non-hispanic ethnicity **d)** if they did not have a psychiatric comorbidity, **e)** if they did not have a court/criminal referral to treatment. In this analysis (e.g., Fig. 1). There were no conditions under which females had significantly greater %NMOU than males.

3.2.3 Multivariable logistic regression of NMOU as the outcome: Adjusted odds ratios of NMOU are in Figure 2 and Supplement Table S3 (which also includes data for census divisions). **Overall regression performance:** The regression had an area under the receiver operating curve (AUROC)=74.2 [95%CI 73.3–54.2; $p<0.0001$], Nagelkerke $R^2=0.22$, and a non-significant Hosmer-Lemeshow test. Several variables and conditions had significant odds ratios of the NMOU outcome, summarized below:

Sex: Males had a greater odds of NMOU compared to females, with an adjusted odds ratio of 1.14 ($p=0.0039$). **Length of stay in treatment:** Shorter lengths of the lengths of stay were associated with greater odds of NMOU, compared to the reference category (≥ 366 days). The greatest odds ratios were observed at the shortest lengths of stay (e.g., 31–45 and 46–60 days), and gradually declined with progressively longer lengths of stay. **Route of NMOU:** Those with inhalation, smoking, or injection routes of administration had greater odds of NMOU, compared to those with the oral route. **Age:** Persons in the age range 40–49 had increased odds of NMOU, compared to the reference category (≥ 50). **Housing:** Persons who were unhoused had increased odds of NMOU, compared to those in independent living, whereas those in “dependent” living (e.g., supportive housing) had decreased odds. **Employment:** Those who worked part-time, were unemployed or not in the labor force had greater odds of NMOU, compared to those employed full-time. **Racial background:**

Persons in the African-American racial category had greater odds of NMOU, compared to persons in the white category. **Ethnicity:** Persons of hispanic/latino ethnicity also had increased odds of NMOU, compared to non-hispanic/latinos. **Psychiatric comorbidity:** Persons with psychiatric comorbidities had lower odds of NMOU, compared to those without such comorbidity. **Court or criminal referral to treatment:** Persons referred to treatment through court/criminal referral had lower odds of NMOU, compared to those referred through other means (e.g., self-referral or medical). **Census divisions:** Compared to the reference category (Middle Atlantic), those in the Pacific and South Atlantic divisions had greater odds of NMOU, whereas those in New England, East North Central, East South Central and Mountain had lower odds (see Table S3).

3.3 Sensitivity analyses:

a) Adding relevant interaction terms (e.g., sex by race) to the multivariable regression did not result in an increased AUROC (not shown). **b)** In the main analysis above, we combined episodes with *any* opioid, by cumulating available categories (*heroin, illicit methadone, and other opiates and synthetics*). In a sensitivity analysis, we examined only episodes with *heroin* as the primary substance. The sex disparity in outcomes was also observable in this subset (not shown). **c)** Polysubstance use is an important feature of opioid use disorder [16,55,56]. As a follow-up, we examined the subset with an opioid as primary substance *and* either cocaine or methamphetamine as secondary substance. In a univariate analysis, this subset also exhibited a similar sex disparity for NMOU (not shown).

Discussion:

This is one of the first national community clinic samples examined for NMOU, a major undesirable outcome, prior to discharge from outpatient treatment with MOUD. Overall, males had greater adjusted odds of NMOU than females. The magnitude of the overall sex difference is relatively small, but survived adjustment for major demographics and social determinants. This study also shows specific conditions under which disparities in NMOU are observable, including some sex-based disparities.

4.1 Length of stay:

Shorter lengths of stay in treatment were associated with greater odds of NMOU, with the largest odds ratios at lengths of stay of 31–45 days, and gradually declining across longer time in treatment. This adds to prior data indicating that longer duration of MOUD is associated with improved clinical outcomes [44,57]. Future studies could focus on further stratification of length of stay, to focus more directly on potentially different underlying processes in the early stages of MOUD, versus longer term maintenance.

4.2 Route of NMOU:

Overall, persons for whom the main route of use was by inhalation (i.e., snorting), smoking (e.g., “chasing the dragon”) or injection had higher odds of NMOU compared to those using by the oral route. Inhaled opioids cause less potent effects than after i.v. injection, likely due to pharmacokinetic mechanisms [58–61]. Increased frequency of inhalation

and smoking routes has recently been reported [39,42,62], especially for relatively potent synthetic opioids including fentanyl [63–66].

4.3 Age at admission:

Persons admitted to treatment in the age bin 40–49 had greater odds of NMOU, compared to the reference category (age ≥ 50). While some studies report that older patients in therapy with MOUD have better outcomes [67,68], this study focused specifically on *first* episodes of treatment, and it is possible that persons who enter their *first* episode of therapy with MOUD in middle age have sub-optimal outcomes.

4.4 Housing and employment:

Unhoused persons had greater odds of NMOU, compared to those in independent housing, possibly caused by diverse stressors and difficulty in treatment adherence [69,70]. Conversely, persons in “dependent” (e.g., supportive) housing had decreased odds of NMOU, compared to those in independent housing, suggesting that “wraparound” support is associated with better outcomes [71,72]. Persons who worked part-time, unemployed or not in the labor force had greater odds of NMOU, compared to those who worked full time overall. A sex disparity (greater NMOU in males vs females) was observed among those who worked part-time or were unemployed.

4.4.1 Racial background and ethnicity: Persons in the African-American/black category had approximately 2-fold adjusted odds ratio of NMOU, compared to those in the white category. Also, hispanic/latinos had greater odds, compared to non-hispanic/latinos. Race and ethnicity can be considered social determinants of health, which can affect access to MOUD [73], and may also affect the likelihood of being prescribed a specific medication (e.g., methadone versus buprenorphine) [74–76].

4.5 Psychiatric comorbidities:

Psychiatric comorbidities are relatively frequent in individuals with opioid use disorders [77–79], and there may be bidirectional causal links between psychiatric comorbidities and the trajectory of opioid use disorder [80–82]. Males in this study had lower odds of psychiatric comorbidity compared to females, consistent with prior studies [13,83]. Also, presence of a psychiatric comorbidity was associated with *decreased* odds of NMOU, consistent with some, but not all, prior studies [78,84–87]. It may be hypothesized that being in MOUD therapy is associated with broader biopsychosocial support and psychiatric care [88], which may then underlie relatively improved outcomes [82]. Also, mu-, kappa-, and delta- opioid receptor systems (at which the MOUD methadone, buprenorphine and naltrexone have differential pharmacological profiles) are involved in depression, anxiety and stress adaptation mechanisms [89–93]. Due to the coding in TEDS-D, data cannot be further stratified into specific MOUD and psychiatric diagnoses. Therefore, future large-scale studies should examine whether specific comorbid diagnoses are associated with differential outcomes for specific types of MOUD, in a sex-related manner.

4.6 Court or criminal justice referral:

Persons with a court/criminal justice referral had decreased odds of NMOU, compared the larger group without such a referral. Some prior studies indicate that persons with court/criminal justice referrals have relatively improved clinical outcomes with MOUD [94–96]. Intriguingly, there was no sex disparity in outcome among those with a court/criminal justice referral, while there were greater odds of NMOU in males vs females, in the larger category referred to treatment through other means (e.g., self-referral or medical).

4.7 General limitations:

Some limitations of the study should be considered: **a)** The “gender” variable in TEDS-D is coded as male/female (i.e., typical definitions of sex); therefore the relative influences of sex and gender *per se* cannot be separated herein. Future large scale investigation of gender on outcomes of MOUD therapy is needed [97,98]. **b)** There is potential selection bias, based on publicly-funded clinics that contribute data to TEDS-D, compared to providers from other settings [99]. **c)** This study did not examine baseline differences in clinical severity or trajectory prior to treatment entry. It is possible that these also differ on the basis of sex or (or other covariates studied herein), and that this affects the NMOU outcome. **d)** As mentioned above, further detail on the specific MOUD is not available, nor on its duration or adherence during the treatment episode. **e)** Future studies could extend these analyses to persons who are in other racial categories, other than African-American or white [100]. **f)** In general, observational studies such as this cannot be used to discern causality of relationships between predictor variables and the measured outcome.

4.8 Conclusions and future studies:

Based on these findings and recent trends [66], enhanced focus on persons using NMOU by the smoking and inhalation routes should be prioritized [62,66], in order to better understand their personalized needs for MOUD care. Also, interventions and implementation strategies focusing on strengthening MOUD care for persons with multiple and potentially intersecting vulnerabilities (e.g., based on social determinants including race, employment, and housing status) are also important [101,102]. While this study indicates that females, once in MOUD treatment, have slightly lower odds of NMOU in adjusted analysis, prior studies show that their *accessibility* to treatment is a barrier [10]. Therefore, future interventions enhancing treatment access by females should be examined, including focus on gender-related features and stigma [103,104]. More broadly, large-scale studies should examine multiple potential mechanisms that may contribute to the observed sex disparities in outcomes of MOUD and in the trajectory of OUD overall [1], including differences in clinical severity, specific comorbid processes [105,106], and neurocognitive features [7,107–109].

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Abbreviations:

95%CI	95% confidence intervals
AUROC	Area under the receiver-operating curve
df	degrees of freedom
LOS	Length of service (duration of treatment episode)
M/F	male / female ratio
MOUD	medications for opioid use disorder
NMOU	non-medical opioid use in the month prior to discharge (binary outcome under study)
NS	Not significant
Ref	Reference category for multivariable logistic regression
TEDS-D	Treatment episode data set - Discharges
MOR	mu-opioid receptor

References

1. Butelman ER, Huang Y, Epstein DH, Shaham Y, Goldstein RZ, Volkow ND, et al. Overdose mortality rates for opioids and stimulant drugs are substantially higher in men than in women: state-level analysis. *Neuropsychopharmacology*. 2023. doi:10.1038/s41386-023-01601-8
2. Wilson N, Kariisa M, Seth P, Smith H 4th, Davis NL. Drug and Opioid-Involved Overdose Deaths - United States, 2017–2018. *MMWR Morb Mortal Wkly Rep*. 2020;69: 290–297. doi:10.15585/mmwr.mm6911a4 [PubMed: 32191688]
3. Towers EB, Williams IL, Qillawala EI, Rissman EF, Lynch WJ. Sex/Gender Differences in the Time-Course for the Development of Substance Use Disorder: A Focus on the Telescoping Effect. *Pharmacol Rev*. 2023;75: 217–249. doi:10.1124/pharmrev.121.000361 [PubMed: 36781217]
4. Little KM, Kosten TA. Focus on Fentanyl in Females: Sex and Gender Differences in the Physiological and Behavioral Effects of Fentanyl. *Front Neuroendocrinol*. 2023; 101096. doi:10.1016/j.yfrne.2023.101096
5. McEvoy A, Chawar C, Lamri A, Hudson J, Minuzzi L, Marsh DC, et al. A genome-wide association, polygenic risk score and sex study on opioid use disorder treatment outcomes. *Sci Rep*. 2023;13: 22360. doi:10.1038/s41598-023-49605-0 [PubMed: 38102185]
6. Apsley HB, Brant K, Brothers S, Harrison E, Skogseth E, Schwartz RP, et al. Pregnancy- and parenting-related barriers to receiving medication for opioid use disorder: A multi-paneled qualitative study of women in treatment, women who terminated treatment, and the professionals who serve them. *Womens Health*. 2024;20: 17455057231224181. doi:10.1177/17455057231224181
7. Huang Y, Ceceli AO, Kronberg G, King S, Malaker P, Parvaz MA, et al. Association of Cortico-Striatal Engagement During Cue Reactivity, Reappraisal, and Savoring of Drug and Non-Drug Stimuli With Craving in Heroin Addiction. *Am J Psychiatry*. 2023; appiajp20220759. doi:10.1176/appi.ajp.20220759

8. Kreek MJ, Reed B, Butelman ER. Current status of opioid addiction treatment and related preclinical research. *Sci Adv*. 2019;5: eaax9140. doi:10.1126/sciadv.aax9140
9. Strang J, Volkow ND, Degenhardt L, Hickman M, Johnson K, Koob GF, et al. Opioid use disorder. *Nat Rev Dis Primers*. 2020;6: 3. doi:10.1038/s41572-019-0137-5
10. Marsh JC, Amaro H, Kong Y, Khachikian T, Guerrero E. Gender disparities in access and retention in outpatient methadone treatment for opioid use disorder in low-income urban communities. *J Subst Abuse Treat*. 2021;127: 108399. doi:10.1016/j.jsat.2021.108399
11. Bawor M, Dennis BB, Bhalerao A, Plater C, Worster A, Varenbut M, et al. Sex differences in outcomes of methadone maintenance treatment for opioid use disorder: a systematic review and meta-analysis. *CMAJ Open*. 2015;3: E344–51. doi:10.9778/cmajo.20140089
12. Zheng W, Cavrak M, Bowles H, Deng Y, Wen S, Gao S, et al. 10-year retention of a comprehensive treatment model of buprenorphine for opioid use disorder. *J Addict Dis*. 2024; 1–8. doi:10.1080/10550887.2024.2315366
13. Huhn AS, Berry MS, Dunn KE. Review: Sex-Based Differences in Treatment Outcomes for Persons With Opioid Use Disorder. *Am J Addict*. 2019;28: 246–261. doi:10.1111/ajad.12921 [PubMed: 31131505]
14. Vigna-Taglianti FD, Burrone P, Mathis F, Versino E, Beccaria F, Rotelli M, et al. Gender Differences in Heroin Addiction and Treatment: Results from the VEdeTTE Cohort. *Subst Use Misuse*. 2016;51: 295–309. doi:10.3109/10826084.2015.1108339 [PubMed: 26872763]
15. Weinstein ZM, Kim HW, Cheng DM, Quinn E, Hui D, Labelle CT, et al. Long-term retention in Office Based Opioid Treatment with buprenorphine. *J Subst Abuse Treat*. 2017;74: 65–70. doi:10.1016/j.jsat.2016.12.010 [PubMed: 28132702]
16. Krawczyk N, Williams AR, Saloner B, Cerdá M. Who stays in medication treatment for opioid use disorder? A national study of outpatient specialty treatment settings. *J Subst Abuse Treat*. 2021;126: 108329. doi:10.1016/j.jsat.2021.108329
17. Hallowell BD, Chambers LC, Samuels EA, Bratberg J, McDonald J, Nitenson A, et al. Sociodemographic and prescribing characteristics that impact long-term retention in buprenorphine treatment for opioid use disorder among a statewide population. *Drug Alcohol Depend*. 2022;241: 109680. doi:10.1016/j.drugalcdep.2022.109680
18. Vo HT, Robbins E, Westwood M, Lezama D, Fishman M. Relapse Prevention Medications in Community Treatment for Young Adults with Opioid Addiction. *Subst Abus*. 2016; 0. doi:10.1080/08897077.2016.1143435
19. Sayre SL, Schmitz JM, Stotts AL, Averill PM, Rhoades HM, Grabowski JJ. Determining predictors of attrition in an outpatient substance abuse program. *Am J Drug Alcohol Abuse*. 2002;28: 55–72. doi:10.1081/ada-120001281 [PubMed: 11853135]
20. Levine AR, Lundahl LH, Ledgerwood DM, Lisieski M, Rhodes GL, Greenwald MK. Gender-specific predictors of retention and opioid abstinence during methadone maintenance treatment. *J Subst Abuse Treat*. 2015;54: 37–43. doi:10.1016/j.jsat.2015.01.009 [PubMed: 25795601]
21. Williams AR, Aronowitz SV, Rowe C, Gallagher R, Behar E, Bisaga A. Telehealth for opioid use disorder: retention as a function of demographics and rurality. *Am J Drug Alcohol Abuse*. 2023;49: 260–265. doi:10.1080/00952990.2023.2180382 [PubMed: 36961998]
22. Rudolph KE, Russell M, Luo SX, Rotrosen J, Nunes EV. Under-representation of key demographic groups in opioid use disorder trials. *Drug Alcohol Depend Rep*. 2022;4. doi:10.1016/j.dadr.2022.100084
23. Eghaneyan BH, Sanchez K, Haeny AM, Montgomery L, Lopez-Castro T, Burlew AK, et al. Hispanic participants in the National Institute on Drug Abuse's Clinical Trials Network: A scoping review of two decades of research. *Addict Behav Rep*. 2020;12: 100287. doi:10.1016/j.abrep.2020.100287
24. Braveman P, Gottlieb L. The social determinants of health: it's time to consider the causes of the causes. *Public Health Rep*. 2014;129 Suppl 2: 19–31. doi:10.1177/00333549141291S206
25. Ross RK, Nunes EV, Olfson M, Shulman M, Krawczyk N, Stuart EA, et al. Comparative effectiveness of extended-release naltrexone and sublingual buprenorphine for treatment of opioid use disorder among Medicaid patients. *Addiction*. 2024;119: 1975–1986. doi:10.1111/add.16630 [PubMed: 39099417]

26. Sheldrick RC. Randomized trials vs real-world evidence: How can both inform decision-making? *JAMA*. 2023;329: 1352–1353. doi:10.1001/jama.2023.4855 [PubMed: 37097366]
27. FDA. Real-World Evidence. In: Food and Drug Administration [Internet]. Available: <https://www.fda.gov/science-research/science-and-research-special-topics/real-world-evidence>
28. Klonoff DC. The expanding role of real-world evidence trials in health care decision making. *J Diabetes Sci Technol*. 2020;14: 174–179. doi:10.1177/1932296819832653 [PubMed: 30841738]
29. Durand L, Boland F, O'Driscoll D, Bennett K, Barry J, Keenan E, et al. Factors associated with early and later dropout from methadone maintenance treatment in specialist addiction clinics: a six-year cohort study using proportional hazards frailty models for recurrent treatment episodes. *Drug Alcohol Depend*. 2021;219: 108466. doi:10.1016/j.drugalcdep.2020.108466
30. Franz B, Cronin CE, Lindenfeld Z, Pagan JA, Lai A, Krawczyk N, et al. Rural-urban disparities in the availability of hospital-based screening, medications for opioid use disorder, and addiction consult services. *J Subst Use Addict Treat*. 2023; 209280. doi:10.1016/j.josat.2023.209280
31. Mennis J, Stahler GJ, El Magd SA, Baron DA. How long does it take to complete outpatient substance use disorder treatment? Disparities among Blacks, Hispanics, and Whites in the US. *Addict Behav*. 2019;93: 158–165. doi:10.1016/j.addbeh.2019.01.041 [PubMed: 30711669]
32. Askari MS, Martins SS, Mauro PM. Medication for opioid use disorder treatment and specialty outpatient substance use treatment outcomes: Differences in retention and completion among opioid-related discharges in 2016. *J Subst Abuse Treat*. 2020;114: 108028. doi:10.1016/j.josat.2020.108028
33. Stenersen MR, Thomas K, Struble C, Moore KE, Burke C, McKee S. Termination From Substance Use Disorder Treatment in the United States: Residential and Outpatient Settings. *J Stud Alcohol Drugs*. 2023;84: 476–484. doi:10.15288/jsad.21-00221 [PubMed: 36971734]
34. Goldstein RZ. Neuropsychosocial Measures as Alternatives to Drug Use Outcomes in Clinical Trials for Addiction. *JAMA Psychiatry*. 2022 [cited 27 Jul 2022]. doi:10.1001/jamapsychiatry.2022.1970
35. Kowalczyk WJ, Phillips KA, Jobes ML, Kennedy AP, Ghitza UE, Agage DA, et al. Clonidine Maintenance Prolongs Opioid Abstinence and Decouples Stress From Craving in Daily Life: A Randomized Controlled Trial With Ecological Momentary Assessment. *Am J Psychiatry*. 2015;172: 760–767. doi:10.1176/appi.ajp.2014.14081014 [PubMed: 25783757]
36. Burgess-Hull AJ, Panlilio LV, Preston KL, Epstein DH. Trajectories of Craving During Medication-Assisted Treatment for Opioid-Use Disorder: Subtyping for Early Identification of Higher Risk. *Drug Alcohol Depend*. 2022; 109362. doi:10.1016/j.drugalcdep.2022.109362
37. Stahler GJ, Mennis J. Treatment outcome disparities for opioid users: Are there racial and ethnic differences in treatment completion across large US metropolitan areas? *Drug Alcohol Depend*. 2018;190: 170–178. doi:10.1016/j.drugalcdep.2018.06.006 [PubMed: 30041092]
38. Surratt H, Kurtz SP, Cicero TJ. Alternate routes of administration and risk for HIV among prescription opioid abusers. *J Addict Dis*. 2011;30: 334–341. doi:10.1080/10550887.2011.609805 [PubMed: 22026525]
39. Fischer B, Robinson T, Jutras-Aswad D. Three noteworthy idiosyncrasies related to Canada's opioid-death crisis, and implications for public health-oriented interventions. *Drug Alcohol Rev*. 2023. doi:10.1111/dar.13796
40. Volkow ND. Collision of the COVID-19 and Addiction Epidemics. *Ann Intern Med*. 2020;173: 61–62. doi:10.7326/m20-1212 [PubMed: 32240293]
41. SAMHSA, Substance Abuse and Mental Health Services Administration. TREATMENT EPISODE DATA SET DISCHARGES (TEDS-D) 2019 Codebook. 2021. Available: https://www.datafiles.samhsa.gov/sites/default/files/field-uploads-protected/studies/TEDS-D-2019/TEDS-D-2019-datasets/TEDS-D-2019-DS0001/TEDS-D-2019-DS0001-info/TEDS-D-2019-DS0001-infocodebook_V1.pdf
42. Huhn AS, Strain EC, Tompkins DA, Dunn KE. A hidden aspect of the U.S. opioid crisis: Rise in first-time treatment admissions for older adults with opioid use disorder. *Drug Alcohol Depend*. 2018;193: 142–147. doi:10.1016/j.drugalcdep.2018.10.002 [PubMed: 30384321]
43. Kitsantas P, Gimm G, Aljoudi SM. Treatment outcomes among pregnant women with cannabis use disorder. *Addict Behav*. 2023;144: 107723. doi:10.1016/j.addbeh.2023.107723

44. Wakeman SE, Larochelle MR, Ameli O, Chaisson CE, McPheeters JT, Crown WH, et al. Comparative Effectiveness of Different Treatment Pathways for Opioid Use Disorder. *JAMA Netw Open*. 2020;3: e1920622. doi:10.1001/jamanetworkopen.2019.20622
45. Ross RK, Nunes EV, Olfson M, Shulman M, Krawczyk N, Stuart EA, et al. Comparative effectiveness of extended release naltrexone and sublingual buprenorphine for treatment of opioid use disorder among Medicaid patients. *medRxiv*. 2024. doi:10.1101/2024.01.24.24301555
46. Blanco C, Volkow ND. Management of opioid use disorder in the USA: present status and future directions. *Lancet*. 2019;393: 1760–1772. doi:10.1016/S0140-6736(18)33078-2 [PubMed: 30878228]
47. Zamora TI, Padilla AM. Making sense of conflicting messages of multiracial identity: a systematic review. *Front Psychol*. 2024;15: 1307624. doi:10.3389/fpsyg.2024.1307624
48. Garcia L, Follis S, Thomson CA, Breathett K, Cené CW, Jimenez M, et al. Taking action to advance the study of race and ethnicity: the Women's Health Initiative (WHI). *Womens Midlife Health*. 2022;8: 1. doi:10.1186/s40695-021-00071-6 [PubMed: 34983682]
49. US Census Bureau. Research to improve data on race and ethnicity. 2022 [cited 10 Sep 2024]. Available: <https://www.census.gov/about/our-research/race-ethnicity.html>
50. Brown LD, Tony Cai T, DasGupta A. Interval Estimation for a Binomial Proportion. *SSO Schweiz Monatsschr Zahnheilkd*. 2001;16: 101–133. doi:10.1214/ss/1009213286
51. Hosmer DW Jr, Lemeshow S. *Applied Logistic Regression*. John Wiley & Sons; 2004. Available: <https://play.google.com/store/books/details?id=Po0RLQ7USIMC>
52. Osborne JW. *Best Practices in Logistic Regression*. SAGE Publications; 2014. Available: <https://play.google.com/store/books/details?id=wUQBwQEACAAJ>
53. Meurer WJ, Tolles J. Logistic regression diagnostics: Understanding how well a model predicts outcomes. *JAMA*. 2017;317: 1068–1069. doi:10.1001/jama.2016.20441 [PubMed: 28291878]
54. Hosmer DW, Lemeshow S. Goodness of fit tests for the multiple logistic regression model. *Communications in Statistics - Theory and Methods*. 1980;9: 1043–1069. doi:10.1080/03610928008827941
55. Butelman ER, Chen CY, Brown KG, Kreek MJ. Escalation of drug use in persons dually diagnosed with opioid and cocaine dependence: Gender comparison and dimensional predictors. *Drug Alcohol Depend*. 2019;205: 107657. doi:10.1016/j.drugalcdep.2019.107657
56. Luo SX, Feaster DJ, Liu Y, Balise RR, Hu M-C, Bouzoubaa L, et al. Individual-Level Risk Prediction of Return to Use During Opioid Use Disorder Treatment. *JAMA Psychiatry*. 2023. doi:10.1001/jamapsychiatry.2023.3596
57. Kimber J, Copeland L, Hickman M, Macleod J, McKenzie J, De Angelis D, et al. Survival and cessation in injecting drug users: prospective observational study of outcomes and effect of opiate substitution treatment. *BMJ*. 2010;341: c3172. doi:10.1136/bmj.c3172 [PubMed: 20595255]
58. Jones JD, Madera G, Comer SD. The reinforcing and subjective effects of intravenous and intranasal buprenorphine in heroin users. *Pharmacol Biochem Behav*. 2014;122: 299–306. doi:10.1016/j.pbb.2014.04.012 [PubMed: 24793093]
59. Lofwall MR, Moody DE, Fang WB, Nuzzo PA, Walsh SL. Pharmacokinetics of intranasal crushed OxyContin and intravenous oxycodone in nondependent prescription opioid abusers. *J Clin Pharmacol*. 2012;52: 600–606. doi:10.1177/0091270011401620 [PubMed: 21610203]
60. Foster D, Upton R, Christrup L, Popper L. Pharmacokinetics and pharmacodynamics of intranasal versus intravenous fentanyl in patients with pain after oral surgery. *Ann Pharmacother*. 2008;42: 1380–1387. doi:10.1345/aph.1L168 [PubMed: 18728103]
61. Comer SD, Collins ED, MacArthur RB, Fischman MW. Comparison of intravenous and intranasal heroin self-administration by morphine-maintained humans. *Psychopharmacology*. 1999;143: 327–338. doi:10.1007/s002130050956 [PubMed: 10367549]
62. Tanz LJ, Gladden RM, Dinwiddie AT, Miller KD, Broz D, Spector E, et al. Routes of Drug Use Among Drug Overdose Deaths - United States, 2020–2022. *MMWR Morb Mortal Wkly Rep*. 2024;73: 124–130. doi:10.15585/mmwr.mm7306a2 [PubMed: 38358969]
63. Rosenblum D, Unick J, Ciccarone D. The Rapidly Changing US Illicit Drug Market and the Potential for an Improved Early Warning System: Evidence from Ohio Drug Crime Labs. *Drug Alcohol Depend*. 2020;208: 107779. doi:10.1016/j.drugalcdep.2019.107779

64. Palamar JJ, Cottler LB, Goldberger BA, Severtson SG, Grundy DJ, Iwanicki JL, et al. Trends in characteristics of fentanyl-related poisonings in the United States, 2015–2021. *Am J Drug Alcohol Abuse*. 2022;48: 471–480. doi:10.1080/00952990.2022.2081923 [PubMed: 35704785]
65. Ciccarone D. Fentanyl in the US heroin supply: A rapidly changing risk environment. *Int J Drug Policy*. 2017;46: 107–111. doi:10.1016/j.drugpo.2017.06.010 [PubMed: 28735776]
66. Ciccarone D, Holm N, Ondocsin J, Schlosser A, Fessel J, Cowan A, et al. Innovation and adaptation: The rise of a fentanyl smoking culture in San Francisco. *PLoS One*. 2024;19: e0303403. doi:10.1371/journal.pone.0303403
67. Rajaratnam R, Sivesind D, Todman M, Roane D, Seewald R. The aging methadone maintenance patient: treatment adjustment, long-term success, and quality of life. *J Opioid Manag*. 2009;5: 27–37. doi:10.5055/jom.2009.0004 [PubMed: 19344046]
68. Carew AM, Comiskey C. Treatment for opioid use and outcomes in older adults: a systematic literature review. *Drug Alcohol Depend*. 2018;182: 48–57. doi:10.1016/j.drugalcdep.2017.10.007 [PubMed: 29136566]
69. Gaeta Gazzola M, Carmichael ID, Christian NJ, Zheng X, Madden LM, Barry DT. A National Study of Homelessness, Social Determinants of Health, and Treatment Engagement Among Outpatient Medication for Opioid Use Disorder-Seeking Individuals in the United States. *Subst Abus*. 2023;44: 62–72. doi:10.1177/08897077231167291 [PubMed: 37226909]
70. Fine DR, Critchley N, Hart K, Joyce A, Sporn N, Gaeta J, et al. “I’m on the Right Path”: Exploring 1-Month Retention in a Homeless-Tailored Outpatient-Based Opioid Treatment Program. *Subst Use Addctn J*. 2024; 29767342231218529. doi:10.1177/29767342231218529
71. McLaughlin MF, Li R, Carrero ND, Bain PA, Chatterjee A. Opioid use disorder treatment for people experiencing homelessness: A scoping review. *Drug Alcohol Depend*. 2021;224: 108717. doi:10.1016/j.drugalcdep.2021.108717
72. Miller-Archie SA, Walters SC, Singh TP, Lim S. Impact of supportive housing on substance use–related health care utilization among homeless persons who are active substance users. *Ann Epidemiol*. 2019;32: 1–6.e1. doi:10.1016/j.annepidem.2019.02.002 [PubMed: 30846275]
73. Barnett ML, Meara E, Lewinson T, Hardy B, Chyn D, Onsando M, et al. Racial Inequality in Receipt of Medications for Opioid Use Disorder. *N Engl J Med*. 2023;388: 1779–1789. doi:10.1056/NEJMsa2212412 [PubMed: 37163624]
74. Hansen HB, Siegel CE, Case BG, Bertollo DN, DiRocco D, Galanter M. Variation in use of buprenorphine and methadone treatment by racial, ethnic, and income characteristics of residential social areas in New York City. *J Behav Health Serv Res*. 2013;40: 367–377. doi:10.1007/s11414-013-9341-3 [PubMed: 23702611]
75. Goedel WC, Shapiro A, Cerdá M, Tsai JW, Hadland SE, Marshall BDL. Association of Racial/Ethnic Segregation With Treatment Capacity for Opioid Use Disorder in Counties in the United States. *JAMA Netw Open*. 2020;3: e203711. doi:10.1001/jamanetworkopen.2020.3711
76. Chhabra N, Smith D, Dickinson G, Caglianone L, Taylor RA, D’Onofrio G, et al. Trends and disparities in initiation of buprenorphine in US emergency departments, 2013–2022. *JAMA Netw Open*. 2024;7: e2435603. doi:10.1001/jamanetworkopen.2024.35603
77. Butelman ER, Bacciardi S, Maremmani AGI, Darst-Campbell M, Correa da Rosa J, Kreek MJ. Can a rapid measure of self-exposure to drugs of abuse provide dimensional information on depression comorbidity? *Am J Addict*. 2017;26: 632–639. doi:10.1111/ajad.12578 [PubMed: 28654734]
78. Gelkopf M, Weizman T, Melamed Y, Adelson M, Bleich A. Does psychiatric comorbidity affect drug abuse treatment outcome? A prospective assessment of drug abuse, treatment tenure and infectious diseases in an Israeli methadone maintenance clinic. *Isr J Psychiatry Relat Sci*. 2006;43: 126–136. Available: <https://www.ncbi.nlm.nih.gov/pubmed/16910375> [PubMed: 16910375]
79. Substance Abuse and Mental Health Services Administration (SAMHSA)}. Co-Occurring Mental Health and Substance Use Services. 2024;Publication No. PEP24–01-008. Available: <http://store.samhsa.gov>
80. Rosoff DB, Smith GD, Lohoff FW. Prescription Opioid Use and Risk for Major Depressive Disorder and Anxiety and Stress-Related Disorders: A Multivariable Mendelian Randomization Analysis. *JAMA Psychiatry*. 2020 [cited 12 Nov 2020]. doi:10.1001/jamapsychiatry.2020.3554

81. Rogers AH, Zvolensky MJ, Ditre JW, Buckner JD, Asmundson GJG. Association of opioid misuse with anxiety and depression: A systematic review of the literature. *Clin Psychol Rev.* 2021;84: 101978. doi:10.1016/j.cpr.2021.101978
82. Martins SS, Keyes KM, Storr CL, Zhu H, Chilcoat HD. Pathways between nonmedical opioid use/dependence and psychiatric disorders: results from the National Epidemiologic Survey on Alcohol and Related Conditions. *Drug Alcohol Depend.* 2009;103: 16–24. doi:10.1016/j.drugalcdep.2009.01.019 [PubMed: 19414225]
83. Braciszewski JM, Idu AE, Yarborough BJH, Stumbo SP, Bobb JF, Bradley KA, et al. Sex Differences in Comorbid Mental and Substance Use Disorders Among Primary Care Patients With Opioid Use Disorder. *Psychiatr Serv.* 2022;73: 1330–1337. doi:10.1176/appi.ps.202100665 [PubMed: 35707859]
84. Rosic T, Naji L, Bawor M, Dennis BB, Plater C, Marsh DC, et al. The impact of comorbid psychiatric disorders on methadone maintenance treatment in opioid use disorder: a prospective cohort study. *Neuropsychiatr Dis Treat.* 2017;13: 1399–1408. doi:10.2147/NDT.S129480 [PubMed: 28579787]
85. Zhu Y, Mooney LJ, Yoo C, Evans EA, Kelleghan A, Saxon AJ, et al. Psychiatric comorbidity and treatment outcomes in patients with opioid use disorder: Results from a multisite trial of buprenorphine-naloxone and methadone. *Drug Alcohol Depend.* 2021;228: 108996. doi:10.1016/j.drugalcdep.2021.108996
86. Krawczyk N, Feder KA, Saloner B, Crum RM, Kealhofer M, Mojtai R. The association of psychiatric comorbidity with treatment completion among clients admitted to substance use treatment programs in a U.S. national sample. *Drug Alcohol Depend.* 2017;175: 157–163. doi:10.1016/j.drugalcdep.2017.02.006 [PubMed: 28432939]
87. Choi S, Adams SM, Morse SA, MacMaster S. Gender differences in treatment retention among individuals with co-occurring substance abuse and mental health disorders. *Subst Use Misuse.* 2015;50: 653–663. doi:10.3109/10826084.2014.997828 [PubMed: 25587672]
88. Hammond CJ, Park G, Kady A, Rathod K, Rahman N, Vidal C, et al. Sex-based differences in psychiatric symptoms and opioid abstinence during buprenorphine/naloxone treatment in adolescents with opioid use disorders. *J Subst Abuse Treat.* 2022;133: 108495. doi:10.1016/j.jsat.2021.108495
89. Browne CA, Falcon E, Robinson SA, Berton O, Lucki I. Reversal of Stress-Induced Social Interaction Deficits by Buprenorphine. *Int J Neuropsychopharmacol.* 2018;21: 164–174. doi:10.1093/ijnp/pyx079 [PubMed: 29020387]
90. Jacobson ML, Wulf HA, Browne CA, Lucki I. The kappa opioid receptor antagonist aticaprant reverses behavioral effects from unpredictable chronic mild stress in male mice. *Psychopharmacology.* 2020. doi:10.1007/s00213-020-05649-y
91. Huang P, Tunis J, Parry C, Tallarida R, Liu-Chen LY. Synergistic antidepressant-like effects between a kappa opioid antagonist (LY2444296) and a delta opioid agonist (ADL5859) in the mouse forced swim test. *Eur J Pharmacol.* 2016;781: 53–59. doi:10.1016/j.ejphar.2016.03.061 [PubMed: 27044434]
92. Bidlack JM, Knapp BI, Deaver DR, Plotnikava M, Arnelle D, Wonsey AM, et al. In Vitro Pharmacological Characterization of Buprenorphine, Samidorphan, and Combinations Being Developed as an Adjunctive Treatment of Major Depressive Disorder. *J Pharmacol Exp Ther.* 2018;367: 267–281. doi:10.1124/jpet.118.249839 [PubMed: 30108159]
93. Perrine SA, Hoshaw BA, Unterwald EM. Delta opioid receptor ligands modulate anxiety-like behaviors in the rat. *Br J Pharmacol.* 2006;147: 864–872. doi:10.1038/sj.bjp.0706686 [PubMed: 16491101]
94. Coviello DM, Zanis DA, Wesnoski SA, Palman N, Gur A, Lynch KG, et al. Does mandating offenders to treatment improve completion rates? *J Subst Abuse Treat.* 2013;44: 417–425. doi:10.1016/j.jsat.2012.10.003 [PubMed: 23192219]
95. Lucabeche VX, Quinn PV. Court-Mandated Treatment Outcomes for Prescribed Opioid Use Disorder: A Gender Based Study. *J Drug Issues.* 2022;52: 47–66. doi:10.1177/00220426211044410
96. Stahler GJ, Mennis J, Stein LAR, Belenko S, Rohsenow DJ, Grunwald HE, et al. Treatment outcomes associated with medications for opioid use disorder (MOUD) among criminal justice-

referred admissions to residential treatment in the U.S., 2015–2018. *Drug Alcohol Depend.* 2022;236: 109498. doi:10.1016/j.drugalcdep.2022.109498

97. Paschen-Wolff MM, Kidd JD, Paine EA. The State of the Research on Opioid Outcomes Among Lesbian, Gay, Bisexual, Transgender, Queer, and Other Sexuality- and Gender-Diverse Populations: A Scoping Review. *LGBT Health.* 2023;10: 1–17. doi:10.1089/lgbt.2022.0036 [PubMed: 36318023]
98. Bahji A, Bastien G, Bach P, Choi J, Le Foll B, Lim R, et al. The Association Between Self-Reported Anxiety and Retention in Opioid Agonist Therapy: Findings From a Canadian Pragmatic Trial. *Can J Psychiatry.* 2023; 7067437231194385. doi:10.1177/07067437231194385
99. Xu KY, Gertner AK, Greenfield SF, Williams AR, Grucza RA. Treatment setting and buprenorphine discontinuation: an analysis of multi-state insurance claims. *Addict Sci Clin Pract.* 2024;19: 17. doi:10.1186/s13722-024-00450-0 [PubMed: 38493109]
100. Flanagan A, Frey T, Christiansen SL, AMA Manual of Style Committee. Updated guidance on the reporting of race and ethnicity in medical and science journals. *JAMA.* 2021;326: 621–627. doi:10.1001/jama.2021.13304 [PubMed: 34402850]
101. Williams CT, Latkin CA. Neighborhood socioeconomic status, personal network attributes, and use of heroin and cocaine. *Am J Prev Med.* 2007;32: S203–10. doi:10.1016/j.amepre.2007.02.006 [PubMed: 17543712]
102. Jalali MS, Botticelli M, Hwang RC, Koh HK, McHugh RK. The opioid crisis: a contextual, social-ecological framework. *Health Res Policy Syst.* 2020;18: 87. doi:10.1186/s12961-020-00596-8 [PubMed: 32762700]
103. Gicquelais RE, Genberg BL, Maksut JL, Bohnert ASB, Fernandez AC. Prevalence and correlates of using opioids alone among individuals in a residential treatment program in Michigan: implications for overdose mortality prevention. *Harm Reduct J.* 2022;19: 135. doi:10.1186/s12954-022-00723-4 [PubMed: 36463189]
104. Schiff DM, Work EC, Muftu S, Partridge S, MacMillan KDL, Gray JR, et al. “You have to take this medication, but then you get punished for taking it:” lack of agency, choice, and fear of medications to treat opioid use disorder across the perinatal period. *J Subst Abuse Treat.* 2022;139: 108765. doi:10.1016/j.jsat.2022.108765
105. Wang K, Shafique S, Xiao D, Walter SM, Liu Y, Piamjariyakul U, et al. Repeated measures analysis of opioid use disorder treatment on clinical opiate withdrawal scale in a randomized clinical trial: sex differences. *J Addict Dis.* 2023; 1–12. doi:10.1080/10550887.2022.2131957
106. Wei J, Lambert TY, Valada A, Patel N, Walker K, Lenders J, et al. Single nucleus transcriptomics of ventral midbrain identifies glial activation associated with chronic opioid use disorder. *Nat Commun.* 2023;14: 5610. doi:10.1038/s41467-023-41455-8 [PubMed: 37699936]
107. Weidacker K, Zhao Y, Zhang Y, Whiteford S, Ren Q, Zhang C, et al. Methadone maintenance treatment and impulsivity: premature responding. *J Clin Exp Neuropsychol.* 2023;45: 606–617. doi:10.1080/13803395.2023.2276483 [PubMed: 37916529]
108. de Wit H. Impulsivity as a determinant and consequence of drug use: a review of underlying processes. *Addict Biol.* 2009;14: 22–31. doi:10.1111/j.1369-1600.2008.00129.x [PubMed: 18855805]
109. Gaudreault P-O, King SG, Huang Y, Ceceli AO, Kronberg G, Alia-Klein N, et al. Frontal white matter changes indicate recovery with inpatient treatment in heroin addiction. *bioRxiv.* 2024. doi:10.1101/2024.06.10.24308719

Highlights

- * It is unclear if there are sex-related disparities in outcomes for outpatient therapy with medications for opioid use disorders (MOUD), in large-scale community settings.
- * In this nationally representative study, adult males had significantly greater odds of non-medical opioid use (NMOU) in the month prior to discharge from their first MOUD episode compared to females, adjusting for demographic variables and social determinants of health.
- * Sex disparities in MOUD outcomes occur under specific conditions that can be examined and potentially addressed, with the goal of improving personalized approaches for OUD.

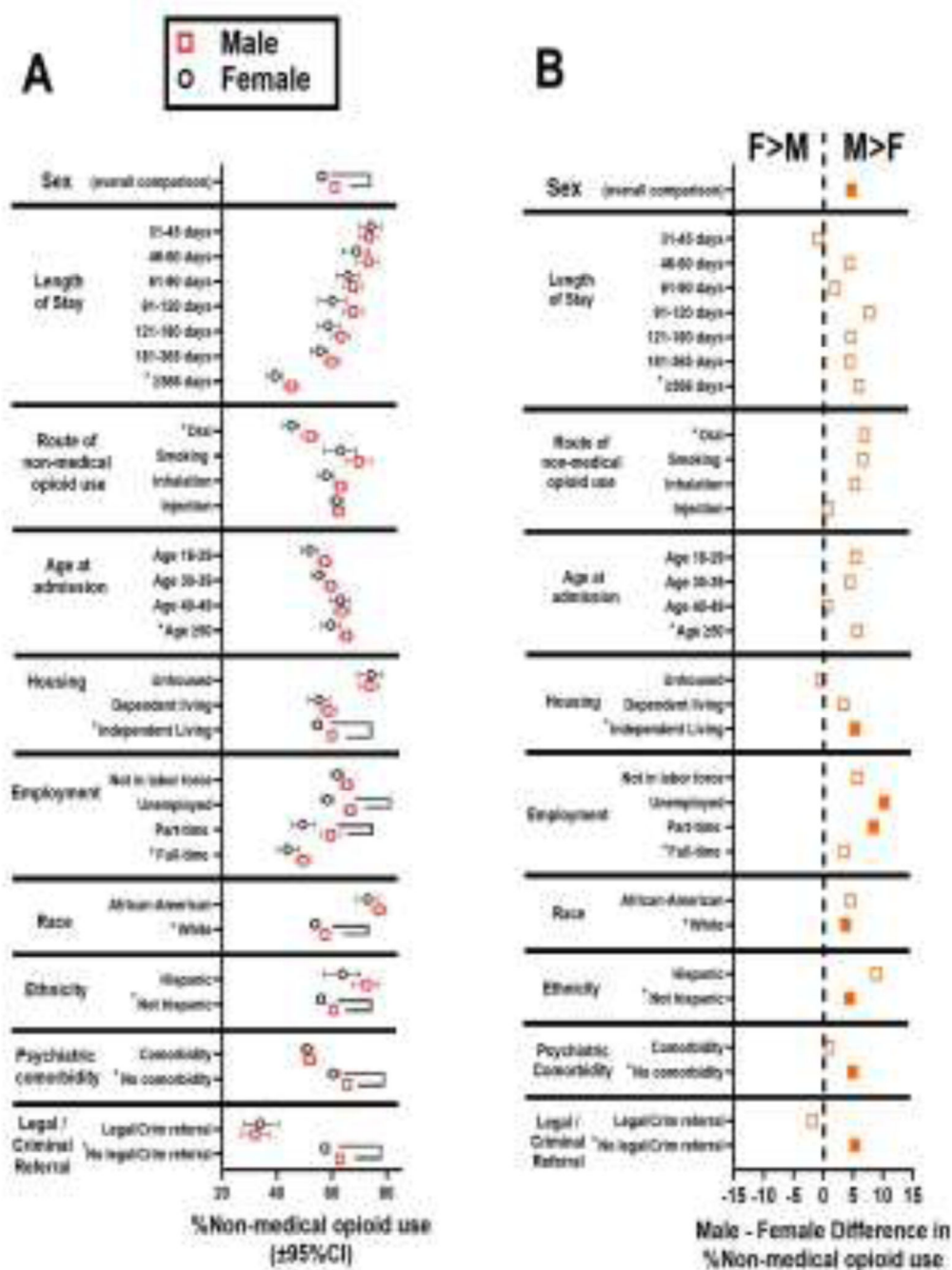


Fig. 1A:

Percent of males and females with NMOU. Brackets indicate conditions with significant 2X2 Chi² analyses by sex, after Bonferroni correction (see Table S2, including data for census divisions). *Indicates the reference condition for multivariable logistic regression (Fig. 2). Fig. 1B: Male-Female differences in %NMOU (from Fig. 1A). Positive values indicate greater %NMOU in males vs females (labeled “M>F” at the top), and negative values indicate the opposite profile (labeled “F>M”). Filled symbols indicate

sex comparisons that survive Bonferroni correction (Table S2); open symbols indicate comparisons that did not reach significance.

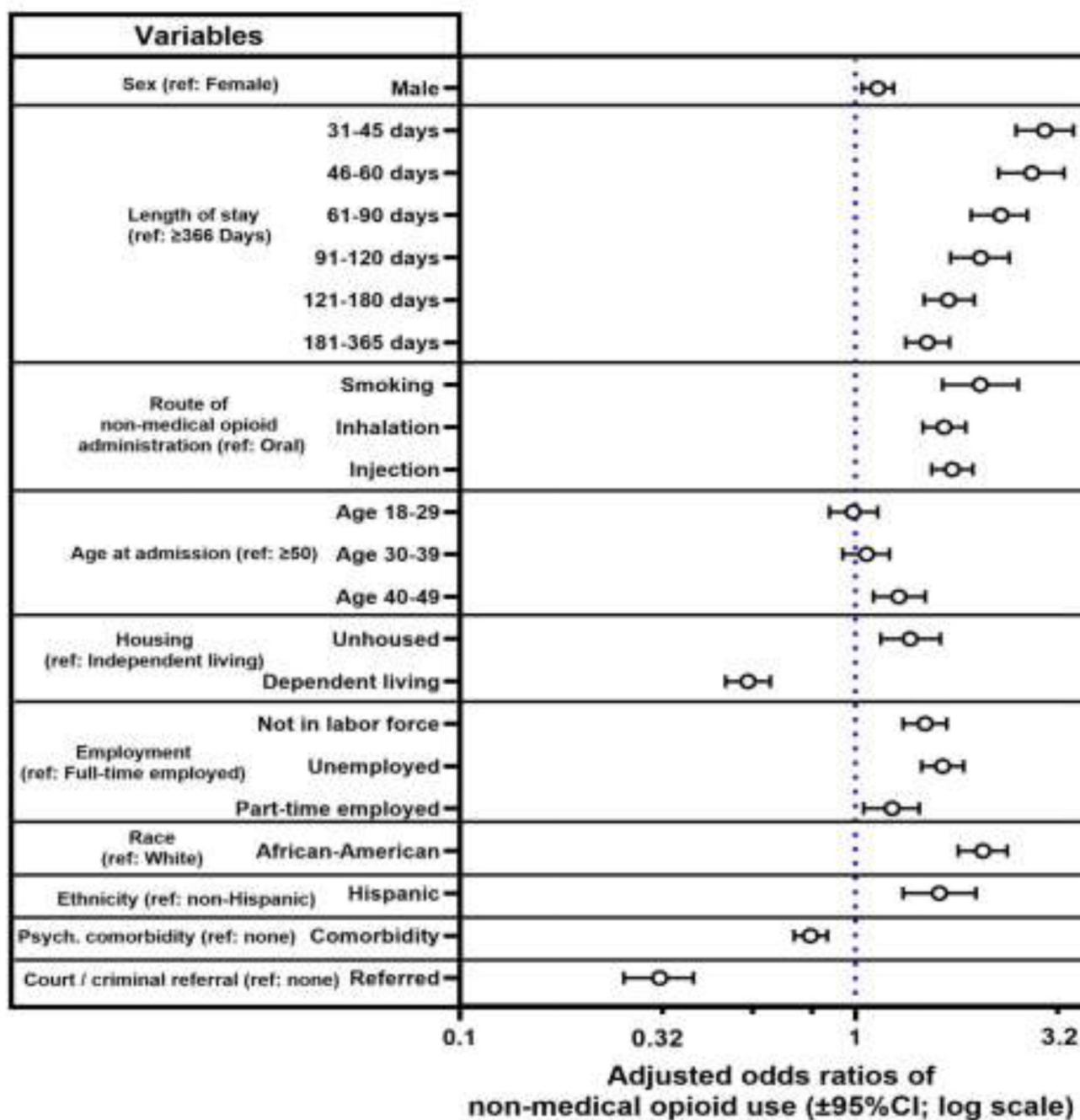


Fig. 2:
Adjusted odds ratios of NMOU, from multivariable logistic regression. Label “ref.” indicates the reference category for each variable. Data for census divisions and full regression parameters are in the Supplement (Table S3).